

AI Voice Agent Design Document

Divyasree Developers – "Whispers of the Wind (WOW)"

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Platform: Vapi AI

LLM: OpenAI GPT-4o Mini

Speech-to-Text: Deepgram Nova-3

Voice AI Framework: Vapi AI

Languages Supported: English & Hindi

Project Type: Outbound AI Voice Lead Qualification Agent

Executive Summary

This project presents an AI-powered outbound voice agent developed for **Divyasree Developers** to automate the first interaction with prospective buyers for the **Whispers of the Wind (WOW)** premium plotted development.

Instead of replacing the sales team, the AI acts as the **first point of contact**, introducing the project, understanding customer requirements, qualifying interest, and transferring qualified prospects to a Property Expert.

The solution demonstrates how Large Language Models (LLMs), real-time speech recognition, and prompt engineering can improve customer engagement while reducing manual effort in lead qualification.

Business Context

Luxury real estate projects receive enquiries from multiple channels including digital advertisements, landing pages, social media campaigns, referral networks, and property portals.

Not every enquiry is sales-ready.

Sales representatives spend considerable time identifying:

- Genuine buyers
- Investment-focused customers
- Budget compatibility
- Purchase readiness

This manual process creates delays and inconsistent customer experiences.

An AI Voice Agent addresses this challenge by conducting the initial qualification automatically before involving a human consultant.

Problem Statement

The current lead qualification process faces several operational challenges:

- High volume of unqualified enquiries
- Repetitive conversations for sales teams
- Inconsistent qualification process
- Delayed response time
- Low productivity
- Higher operational cost

The business requires an intelligent conversational assistant capable of performing structured lead qualification while maintaining a premium customer experience.

Solution Overview

The proposed solution is an outbound AI Voice Agent built on **Vapi AI** using **OpenAI GPT-4o Mini** and **Deepgram** speech recognition.

The AI consultant "Priya" conducts a structured 2–3 minute conversation to:

- Introduce Divyasree Developers
- Ask permission before continuing
- Build rapport
- Understand customer intent
- Evaluate location preference
- Verify budget alignment
- Determine purchase timeline
- Present the project's value proposition
- Schedule a callback with a Property Expert

The conversation is intentionally consultative rather than sales-oriented.

Project Information

Developer: Divyasree Developers

Project: Whispers of the Wind (WOW)

Project Type: Premium Private Valley Villa Plots

Location: Nandi Valley, Near Nandi Hills, North Bengaluru

Plot Size: 1200–3199 sq.ft.

Pricing: ₹92.4 Lakh – ₹2.46 Crore

Possession: December 2029

Business Objectives

The AI Voice Agent was designed with the following objectives:

- Reduce manual lead qualification effort
 - Improve customer response time
 - Maintain consistent communication quality
 - Increase qualified lead conversion
 - Deliver premium customer experience
 - Assist sales consultants rather than replace them
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AI Agent Persona

Agent Name: Priya

Role:

Professional AI Property Consultant

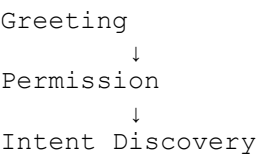
Communication Style:

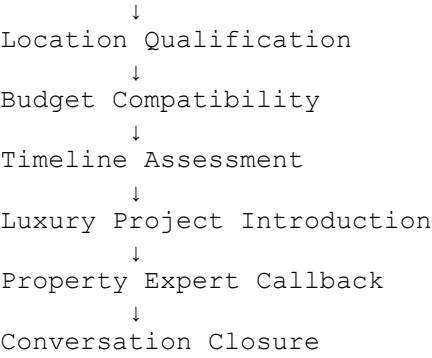
- Warm
- Professional
- Confident
- Respectful
- Conversational
- Premium

The agent avoids aggressive sales behaviour and focuses on understanding customer requirements.

Conversation Design

The conversation follows a deterministic qualification workflow while allowing natural dialogue.





Lead Qualification Framework

The qualification logic is based on four business parameters.

Intent

Identify whether the customer is interested in:

- Self-use
- Investment
- Weekend Home
- General Exploration

Location

Evaluate customer comfort with:

- Nandi Hills
- Devanahalli Corridor

Budget

Rather than directly asking for budget, the AI validates compatibility by stating the project's starting price.

Timeline

Determine purchase readiness:

- Immediate
- 3 Months
- 6 Months
- 1 Year
- Exploring

Prompt Engineering Strategy

The quality of an AI voice agent depends heavily on prompt engineering.

The system prompt was designed using several behavioural constraints.

Persona Definition

The AI consistently behaves as an experienced property consultant.

Conversation Memory

Previously collected customer information is remembered to avoid repeated questions.

Single Question Rule

Only one qualification question is asked at a time.

Premium Tone

The conversation remains polite, concise, and non-intrusive.

Controlled Responses

The AI only answers questions using verified project information.

Knowledge Base

The AI is restricted to verified project information.

Supported topics include:

- Location
- Pricing
- Plot Sizes
- Amenities
- Possession Timeline
- Community Features

If information is unavailable, the AI escalates the query to a Property Expert.

Multilingual Capability

The agent begins every interaction in English.

If the customer requests Hindi, the AI seamlessly switches languages and continues the conversation naturally.

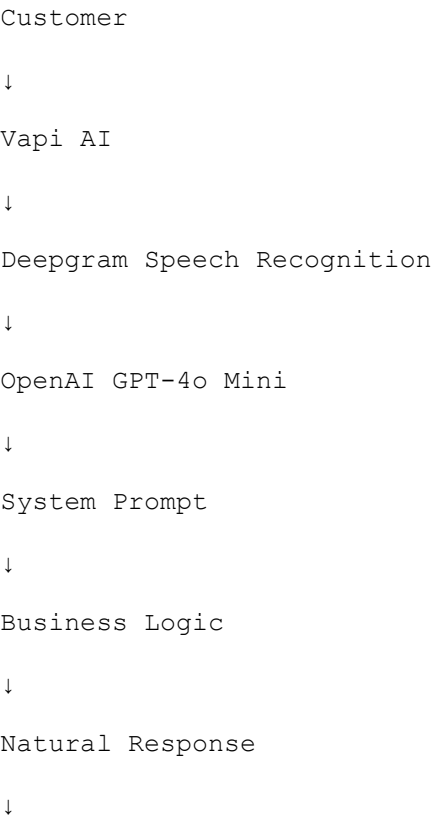
This improves accessibility for customers who prefer regional communication.

Edge Case Handling

The AI was designed to handle several real-world situations.

Scenario	AI Behaviour
Busy Customer	Offer callback
Budget Mismatch	Politely explain price range
Wrong Number	Apologize and end call
Not Interested	Thank and close
Irritated Customer	Remain calm and respectful
Existing Customer	Continue contextually
Hindi Request	Switch language automatically

Technical Architecture



Technology Stack

Component	Technology
Voice Platform	Vapi AI
Language Model	OpenAI GPT-4o Mini
Speech Recognition	Deepgram Nova
Prompting	Custom System Prompt
Language	English + Hindi

Testing

Five representative conversations were recorded.

- 1. Qualified Investor
- 2. Weekend Home Buyer
- 3. Budget Mismatch
- 4. Busy Customer
- 5. Hindi Conversation

Each scenario validates a different branch of the conversation flow.

Business Benefits

Customer Benefits

- Faster responses
- Consistent communication
- Better engagement
- Convenient callback scheduling

Business Benefits

- Reduced manual effort
 - Improved lead qualification
 - Standardized customer interactions
 - Increased sales productivity
 - Better allocation of sales resources
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Future Roadmap

Potential enhancements include:

- CRM integration
- WhatsApp follow-up automation
- Calendar scheduling
- Sentiment analysis
- AI-powered lead scoring
- Real-time analytics dashboard
- Additional regional language support

Conclusion

The AI Voice Agent successfully demonstrates how conversational AI can automate the early stages of luxury real estate lead qualification while maintaining a premium customer experience.

By combining structured conversation design, multilingual support, prompt engineering, and real-time speech technologies, the solution provides an efficient bridge between prospective buyers and human sales consultants.

Rather than replacing human expertise, the AI augments the sales process by identifying qualified prospects, reducing repetitive tasks, and ensuring every customer receives a consistent, professional interaction.